

Why the ancients argued about the arrow and not the sling

The ancient debate on motion is almost entirely rectilinear. Zeno’s arrow, the Vaiśeṣika arrow, Vasubandhu’s denial of passage, Hui Shi’s stick and the Mohist shadow all take straight-line or static cases, while the sling and stone, which is where Newton **defines** centripetal force, is missing from the philosophical argument even though slings were ordinary objects. The absence is systematic, and the reason is that the ancient question and Newton’s are at different rungs of the same ladder: the ancients asked whether motion is possible at all, which is a first-order question about succession, and Newton asked what sustains a departure from straight-line motion, which is second order and becomes askable only once inertia is assumed. Section 3 gives the one ancient case that is genuinely circular, the whirled firebrand, and shows it was used for a third purpose again: neither possibility nor force, but the difference between what appears and what is. Lateral to the main argument; no ledger promotion.

1. Newton’s definition uses the sling

Definition V of the *Principia* introduces the concept with the stone, in the Motte translation held in this repository:

A stone, whirled about in a sling, endeavours to recede from the hand that turns it; and by that endeavour, distends the sling, and that with so much the greater force, as it is revolved with the greater velocity, and as soon as ever it is let go, flies away. That force which opposes itself to this endeavour, and by which the sling perpetually draws back the stone towards the hand, and retains it in its orbit, because it is directed to the hand as the centre of the orbit, I call the centripetal force.

The sentence before it names the planets, “perpetually drawn aside from the rectilinear motions, which otherwise they would pursue”. So the sling is the laboratory model of Proposition I, and the clause “which otherwise they would pursue” is the load-bearing one: the straight line is what happens by default, and only the deviation needs a cause.

2. The ladder, and why the ancients stood on its first rung

Theorem 5 of the paper makes the difference exact. For a signal that is an n -th order departure, with the lower orders unknown and therefore nuisance, the quantity bounded below is $m(P^{(n)})^2 \tau^{2n-1} \gtrsim \kappa$:

Rung	Question	Nuisance	Bound
$n = 1$	Is it moving at all?	position	$mv^2\tau \geq 8z^2\kappa$
$n = 2$	Is a force acting?	position and velocity	$F^2\tau^3/m \geq 18z^2\kappa$

Zeno, the Vaiśeṣika sūtras and Vasubandhu are all at $n = 1$: the thing in dispute is whether the arrow moves, and the unknown to be beaten is merely where it is. The sling is at $n = 2$, and so is Galileo’s comparison and all of Newton’s Book I. The second rung cannot even be **stated** without treating uniform motion as needing no account, because otherwise curved motion is not more puzzling than straight motion and there is no distinguished thing for a centripetal force to explain. Ancient physics on the whole treats rest as the natural state, so the second rung is not available to it, and the case that would exhibit it has no work to do.

The Vaiśeṣika comes closest to opening it. VS 5.1.17 gives impulsion the **first** motion only and hands the rest to the impression the previous motion deposits, so the continuation of motion is already not being charged to a continuing push. What is missing is the further step that a

straight continuation is free while a *curved* one is not, and without that step the sling is an ordinary event rather than a problem.

3. The one circular case, used for a third purpose

Indian sources do have a whirled object: the firebrand circle, *alātacakra*, the ring of fire seen when a burning stick is swung. Vasubandhu uses it in the *Abhidharmakośabhāṣya*, Pradhan 189.23–24, arguing that because contact with the parts is successive, the cognition of a whole is really a cognition of the parts, *avayaveṣv eva tad-buddhir alātacakravat*, “the awareness is of the parts only, as with the firebrand circle”. At Pradhan 33.9 the same image appears with *āśuvṛtṭyā*, “by rapid action”, in asking whether the eye grasps objects of its own size. The firebrand is therefore an argument about **sampling**: a rapid succession presents itself as a continuous extended thing, and the continuity is the observer’s.

That is a third use, and for this programme it is the most interesting of the three. It is the ancient form of the point Theorem 5 makes about records rather than about the world: below the mesh a succession is indistinguishable from the continuous object it mimics, and above it the succession is recoverable. The Buddhist example runs the inference in the direction the theorem does not license, from the appearance being a succession to the whole being unreal, but the phenomenon it names is exactly the one being measured.

4. Consequence for STATE

The case selection in the ancient sources is explained rather than merely noted: the arrow is the first rung and the sling the second, and the second needs inertia before it becomes a question. The Vaiśeṣika impression is the nearest ancient approach to the missing premise. The firebrand supplies the one circular example and uses it for sampling, which is the closest ancient statement of what Theorem 5 says about records. Section 7 of the paper now carries the point in short form.

Open: whether any Greek source argues about the sling rather than merely using it, with Ps.-Aristotle *Mechanica* and Aristotle *Physics* VIII.10 on projectiles as the places to look; and whether the *alātacakra* appears in the Nyāya-Vaiśeṣika commentaries on VS 5.1.16 beside the arrow and the bird, which the *Upaskāra* is reported to do and this collection cannot yet check.